

forceTorqueThrForceMapping

1 Executive Summary

This module maps commanded forces and torques defined in the body frame of the spacecraft to a set of thrusters. It is capable of handling Center of Mass (CoM) offsets and non-controllable axes. In contrast to `thrForceMapping`, this module only handles on-pulsing, but not off-pulsing. Further, it provides a single force/torque projection onto the thrusters and thus lacks some of the robustness features of `thrForceMapping`.

The commanded force and torque input messages are optional, and the associated vectors are zeroed if no input message is connected. This provides a general capability to map control torques, forces, or torques and forces onto a set of thrusters.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for. Both the `cmdTorqueInMsg` and `cmdForceInMsg` are optional.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
<code>cmdTorqueInMsg</code>	<code>CmdTorqueBodyMsgPayload</code>	(optional) vehicle control torque ($\mathbf{L}_r$) input message
<code>cmdForceInMsg</code>	<code>CmdForceBodyMsgPayload</code>	(optional) vehicle control force input message
<code>thrConfigInMsg</code>	<code>THRArrayConfigMsgPayload</code>	thruster cluster configuration input message
<code>vehConfigInMsg</code>	<code>VehicleConfigMsgPayload</code>	vehicle configuration input message
<code>thrForceCmdOutMsg</code>	<code>THRArrayCmdForceMsgPayload</code>	thruster force command output message

3 Module Assumptions and Limitations

This module assumes that each thruster only produces positive thrust (on-pulsing only). The stacked force/torque mapping matrix $[D]$ is assumed to have at least one controllable axis; rows corresponding to completely uncontrollable axes are dropped before the pseudo-inverse is computed. The resulting solution is the minimum-norm least-squares projection onto the thruster set, shifted so the minimum thrust is zero. No off-pulsing, saturation handling, or thruster availability masking is performed within this module.

4 Initialization

The module is configured by:

```
module = forceTorqueThrForceMapping.ForceTorqueThrForceMapping()  
module.modelTag = "forceTorqueThrForceMappingTag"
```

The `cmdForceInMsg` and `cmdTorqueInMsg` are optional; if not connected the corresponding commanded vector is treated as zero. The `thrConfigInMsg` and `vehConfigInMsg` are required:

```
fswSetupThrusters.clearSetup()  
for i in range(numThrusters):  
    fswSetupThrusters.create(rcsLocationData[i], rcsDirectionData[i], maxThrust)
```

```
thrConfigInMsg = fswSetupThrusters.writeConfigMessage()

vehConfigInMsgData = messaging.VehicleConfigMsgPayload()
vehConfigInMsgData.CoM_B = CoM_B
vehConfigInMsg = messaging.VehicleConfigMsg().write(vehConfigInMsgData)
```

The relevant messages must then be subscribed to by the module:

```
module.cmdTorqueInMsg.subscribeTo(cmdTorqueInMsg)
module.cmdForceInMsg.subscribeTo(cmdForceInMsg)
module.thrConfigInMsg.subscribeTo(thrConfigInMsg)
module.vehConfigInMsg.subscribeTo(vehConfigInMsg)
```

5 Detailed Module Description

The following text describes the mathematics behind the `forceTorqueThrForceMapping` module.

5.1 Force and Torque Mapping

The desired force and torque are given as $\mathbf{F}_{req}$ and $\boldsymbol{\tau}_{req}$, respectively. These are both stacked into a single vector

$$\begin{bmatrix} \boldsymbol{\tau}_{req} \\ \mathbf{F}_{req} \end{bmatrix}$$

The i -th thruster position expressed in spacecraft body-fixed coordinates is given by $\mathbf{r}_i$. The unit direction vector of the thruster force is $\hat{\mathbf{g}}_{t_i}$. The thruster force is given as

$$\mathbf{F}_i = F_i \hat{\mathbf{g}}_{t_i}$$

The torque produced by each thruster about the body-fixed CoM is

$$\boldsymbol{\tau}_i = ((\mathbf{r}_i - \mathbf{r}_{COM}) \times \hat{\mathbf{g}}_{t_i}) F_i = \mathbf{d}_i F_i$$

The total force and torque on the spacecraft may be represented as

$$\begin{bmatrix} \boldsymbol{\tau}_{req} \\ \mathbf{F}_{req} \end{bmatrix} = \begin{bmatrix} \mathbf{d}_1 \cdots \mathbf{d}_N \\ \hat{\mathbf{g}}_{t_1} \cdots \hat{\mathbf{g}}_{t_N} \end{bmatrix} \begin{bmatrix} F_1 \\ \vdots \\ F_N \end{bmatrix} = [D] \mathbf{F}$$

The force required by each thruster is computed via the Moore-Penrose pseudo-inverse of $[D]$. Any rows of $[D]$ that contain only zeros (within a small numerical threshold) correspond to uncontrollable axes and are removed beforehand.

$$\mathbf{F} = [D]^T ([D][D]^T)^{-1} \begin{bmatrix} \boldsymbol{\tau}_{req} \\ \mathbf{F}_{req} \end{bmatrix}$$

To ensure no commanded thrust is less than zero, the minimum thrust is subtracted from the thrust vector

$$\mathbf{F} = \mathbf{F} - \min(\mathbf{F})$$

These thrust commands are then written to the output message.

6 Additional Information

The minimum-thrust subtraction in `eq:F_min` guarantees non-negative thruster commands but does not preserve the exact commanded force and torque when the raw pseudo-inverse solution contains negative components. In those cases the system sacrifices some of the requested force/torque fidelity in exchange for producing an on-pulse-only solution. For rank-deficient $[D]$ the minimum-norm solution is returned; for completely uncontrollable command axes the corresponding component is silently dropped.